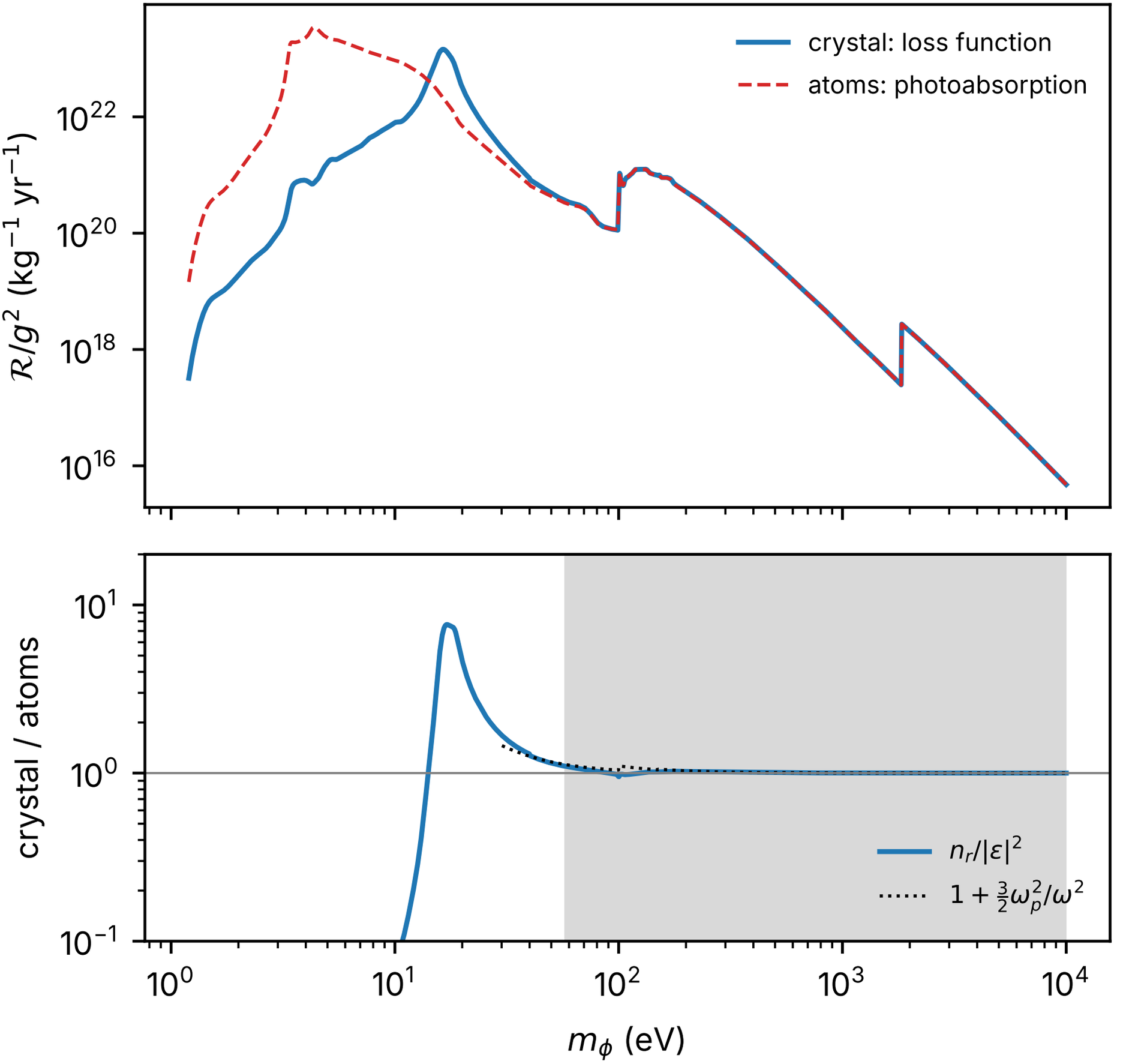
crystal: loss function
atoms: photoabsorption
$\mathcal{R}/g^2$ (kg^{-1} yr^{-1})
10^{22}
10^{20}
10^{18}
10^{16}
crystal / atoms
10^{1}
10^{0}
10^{-1}
$n_r/|\varepsilon|^2$
$1 + \frac{3}{2}\omega_p^2/\omega^2$
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
m_ϕ (eV)